

	$\Delta p = 1.25 \times 10^{-24}$
29.	Ans: B Binding energy of a tritium nucleus $= (\text{mass defect}) c^2$ $= (1.0086 + 2 \times 1.0097 - 3.0160) (1.66 \times 10^{-27}) (3.00 \times 10^8)^2$ $= 1.7928 \times 10^{-12} \text{ J}$ $= 11.2 \text{ MeV}$ In order to separate a tritium nucleus into the individual nucleons, the tritium nucleus will thus need to absorb energy equal or more than 11.2 MeV.
30.	Ans: A